

On the other hand, if you put the entire \$100 in any one asset, either Fund XYZ or Fund ABC, the chance of earning less than \$105 now rises to 37%. This is what I call the sum of the risk. The fundamental law of diversification in action means that the risk of the sum (28%) is less than the sum of the risk (37%).

An additional point to mention is that in Table 3.1, we are dealing with a one-year horizon. It examines the shortfall odds over a 12-month period. What happens if we extend this correlation analysis to a longer time horizon?

Table 3.2 paints the longer-term picture. This time, the returns on money split equally between two investments (in various correlation scenarios) are compared at five years (with a five-year, 5% compounded CD) and at 10 years (with a hypothetical 10-year 5% CD).

**TABLE 3.2 Your Money Is Split Equally Across Two Investments: What Is the Probability of Earning Less Than a 5-Year and a 10-Year 5% Compounded Certificate of Deposit?**

| Correlation Between Investments | 5-Year Probability of Shortfall | 10-Year Probability of Shortfall |
|---------------------------------|---------------------------------|----------------------------------|
| 100%                            | 22.80%                          | 14.59%                           |
| 75%                             | 20.00%                          | 11.70%                           |
| 50%                             | 16.92%                          | 8.79%                            |
| 25%                             | 13.53%                          | 5.96%                            |
| 0%                              | 9.83%                           | 3.39%                            |
| -25%                            | 5.94%                           | 1.37%                            |
| -50%                            | 2.31%                           | 0.24%                            |
| -75%                            | 0.17%                           | 0.00%                            |
| -100%                           | 0.00%                           | 0.00%                            |

E[r]=15%, vol=30%; probability each asset earns <5% compounded for 5 yrs = 22.80%; for 10 yrs = 14.59%

Source: Moshe Milevsky and the IFID Centre, 2008.

As you can see, Table 3.2 has uniformly lower shortfall probabilities than those in Table 3.1. What does that mean? It means that the odds of earning less than a five-year compounded 5% CD decrease the longer you hold the investment.